Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| Title of Proposed Technology | CRISPR-Driven Paper-based Diagnostic Platform for Genetic and Epigenetic Biomarker Detection | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1)] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | *Bio-Medical* | | *Bio-Medical* | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *Medical Bio* | *Reagents / Diagnostic Agents* | *Therapeutic Devices & Diagnostic Devices* | *Bio-signal Measurement / Diagnostic Devices* |
| Concept of Proposed Technology | Rapid and reliable detection of genetic and epigenetic biomarkers from human samples such as blood, urine, and saliva is critical to human health and healthcare organizations.[1,2] There is a need to develop simple, point-of-care (POC for physicians' offices) and point-of-use (POU for patients) devices that are easy to use and lead to very low false positives/negatives.[3-5] Building off our prior experience in CRISPR research[6-8] and its application in diagnostic development,[9-11] we propose a CRISPR-driven, paper-based lateral flow assay (LFA) platform for the rapid detection of genetic and epigenetic biomarkers, including methylated DNA, enabling early diagnosis and personalized intervention. The platform integrates selective nucleic acid recognition by CRISPR/Cas proteins, signal amplification via isothermal amplification methods, and an instrument-free, paper-based visual readout. Upon target recognition, CRISPR effectors generate a robust signal through collateral (trans-cleavage) activity, enabling highly specific detection with minimal false-positive and false-negative | | | |

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| | results. By combining CRISPR-mediated molecular selectivity, isothermal amplification, and capillary-driven lateral flow signal transduction, this platform provides a modular, scalable, and deployable diagnostic solution. Compared to conventional paper-based diagnostics, the proposed system offers enhanced specificity, improved sensitivity for low-abundance biomarkers, and compatibility with POC and POU settings, including physicians' offices and at-home testing. This technology enables rapid, accurate, and accessible biomarker analysis for decentralized healthcare and biomedical research applications. Furthermore, the technology is ripe for commercialization across multiple biomarkers relevant for early-stage diagnostic testing. |
| **Rationale for Joint R&D Project** | We have developed a CRISPR-based immunoassay that leverages the trans-cleavage activity of Cas12a for sensitive chemokine biomarker detection[9-11]. Building on our extensive experience in studying biomolecular interactions using surface plasmon resonance (SPR),[12,13] we are also developing a binding-based detection platform using catalytically deactivated Cas9, Cas12a and Cas13a to enable highly specific biomolecule recognition and detection. However, to advance these technologies toward point-of-care testing (POCT), collaboration with a company that possesses hands-on expertise in designing, constructing, and implementing paper-based platforms capable of multiplex or multimeric detection is particularly essential. Combining our CRISPR detection capabilities with the company's experience in developing multiplexed paper-based systems will enable the creation of robust, deployable diagnostics that integrate high specificity, amplified signals, and practical, user-friendly formats suitable for decentralized testing. A key outcome of the TIPA funding is enhanced intellectual property that builds on existing IP.[11] A longer-term outcome (as follow-on collaboration beyond the scope of this project) is the commercial development of LFA products for POC and POU testing. |
| **Key Words** | 1. Paper-Based Diagnostics<br>2. Lateral flow assay (LFA)<br>3. Clustered Regularly Interspaced Short Palindromic Repeat (CRISPR)<br>4. Epigenetic biomarkers<br>5. Point-of-care testing (POCT) |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level ( 3 ) | End of the Project | Level ( 5 ) |
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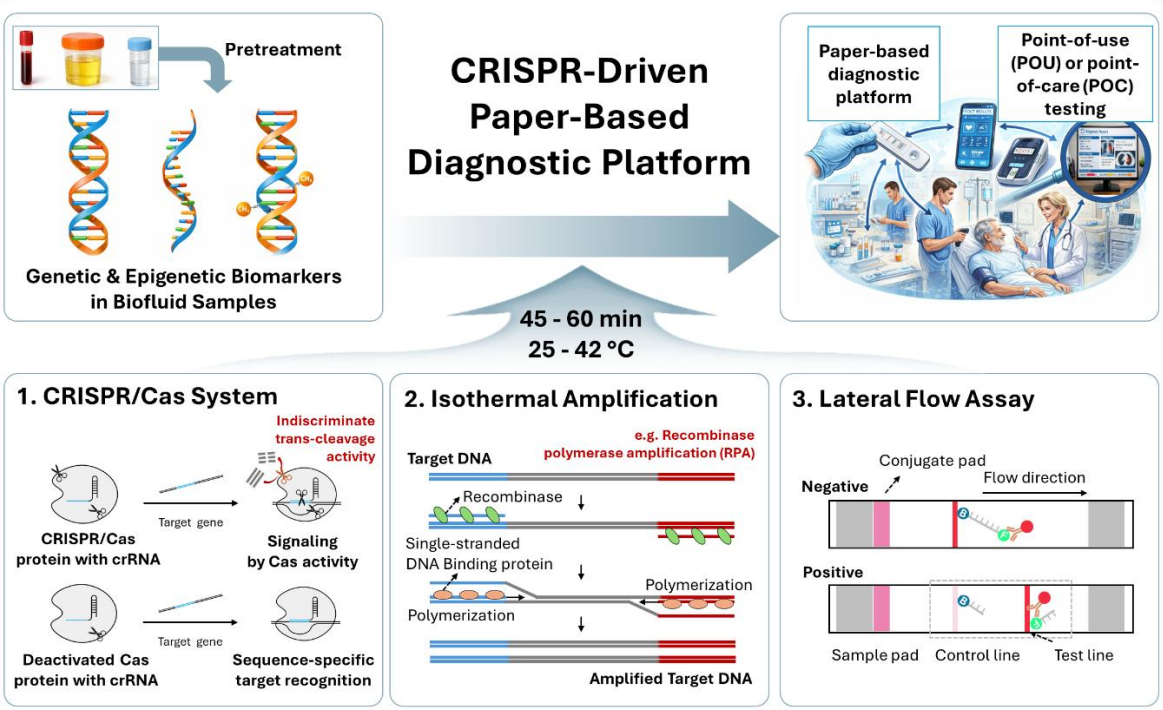
**R&D Objectives**

The goal of this project is to establish a robust CRISPR-based, paper diagnostic platform for rapid, accurate, and accessible detection of diverse genetic and epigenetic biomarkers. By integrating CRISPR/Cas-mediated sequence-specific recognition and signal generation with isothermal nucleic acid amplification on a capillary-driven paper substrate, this research aims to enable sensitive detection of DNA, RNA under instrument-free or minimally instrumented conditions, without the need for temperature cycling or precise thermal control (see **Scheme**). The proposed system is designed to deliver rapid time-to-result and high analytical specificity while supporting early disease diagnosis, real-time epigenetic monitoring, and POC or POC deployment in decentralized healthcare and resource-limited settings (e.g., physician's offices), and for relevant biomarkers, direct POU uses (e.g., patient home testing). These objectives will be achieved through a structured R&D plan consisting of four tasks:

- Task 1. Identification of Genetic Targets and Sample Matrices
- Task 2. Development of CRISPR- and Isothermal Amplification-Based Detection System
- Task 3. System Integration on a Paper-Based LFA Platform
- Task 4. Platform Validation in Simulated Relevant Biological Matrices




**Scheme.** Schematic overview of the CRISPR-driven paper-based diagnostic platform for genetic & epigenic biomarker detection, integrating three core modules: (1) CRISPR/Cas system; (2) Isothermal amplification; and (3) Lateral flow assay (LFA). Portions of the image were created with AI assistance (OpenAI GPT-4).

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| **R&D Target Specifications** | Target performance specifications for this R&D effort are defined to support advancement of a CRISPR-based, paper diagnostic platform from TRL 3 to TRL 5, with emphasis on analytical performance, speed, and operational simplicity for POC and POC applications.<br>**1. Identification of targets**<br>• 5–10 candidate genetic and epigenetic biomarkers, including methylated DNA targets (e.g., post-traumatic stress disorder (PTSD) biomarkers: cg01874867, cg20052079, cg10975897)[14]<br>• ≥ 2 candidate sample matrices (e.g., blood, urine, saliva)<br>**2. Sensitivity**<br>• ≤ 100 copies per reaction for targets using CRISPR/Cas & isothermal amplification steps<br>• ≤ $10^2$–$10^3$ copies per test following integration onto a paper-based LFA<br>**3. Specificity**<br>• ≥ 95% overall specificity<br>• ≥ 95% signal suppression for single-base mismatches<br>• ≤ 5% false negative<br>**4. Validation / benchmarking**<br>• Sensitivity and specificity were benchmarked against standard sequencing and/or Bisulfite sequencing<br>**5. Time-to-result**<br>• ≤ 45-60 minutes total assay time from sample input to readout<br>**6. Operational requirements**<br>• Instrument-free or ≤1 simple heat source, with no requirement for temperature cycling or dynamic temperature changes<br>• Operating temperature range of approximately 20–50 °C<br>**7. Readout**<br>• ≥ Naked-eye detection threshold and/or smartphone-based quantification<br>**8. Multiplexing capability**<br>• ≥ 2–3 targets per LFA strip<br>• ≤ 10% signal cross-reactivity |
| **R&D Plan** | The objective of this R&D effort is to advance a CRISPR-driven, paper-based diagnostic platform from TRL 3 (analytical proof-of-concept) to TRL 5 (component and system validation in a relevant environment). The R&D plan is organized into four technical tasks, with an initial phase covering a 6-month period.<br><br>***Task 1. Initiation of RPI-Korean SME Collaboration, Identification of Genetic Targets and Sample Matrices, and Preparation of Detailed Proposal (Months 0–6, TRL 3)***<br>This task includes the initiation of the collaboration between the PIs at RPI and a Korean SME. During the 6-month initiation period, we will jointly define a focused set of genetic targets and representative sample matrices for technology development. A panel of 5–10 candidate genetic and epigenetic biomarkers will be identified, with target diseases potentially including chronic conditions such as |

cancer, Alzheimer's disease and PTSD, as well as acute conditions like sepsis and organ injury, where early molecular detection is critical. For example, PTSD biomarkers may include cg01874867, cg20052079, and cg10975897.[14] Both unmethylated and methylated DNA targets may be included. In parallel, candidate biological sample matrices, such as blood-, urine-, and sweat-derived environments, will be selected based on their relevance to point-of-care and point-of-use testing.

**Deliverable**: A full proposal prepared jointly by RPI and a Korean SME.

**Milestone**: A minimum of 5 candidate biomarkers in at least two sample matrices identified and mutually agreed upon by RPI and the Korean SME.

***Task 2. Development of CRISPR- and Isothermal Amplification-Based Detection System (Months 7–18, TRL 3–4)***

This task focuses on developing and analytically validating a modular detection system that integrates isothermal amplification with CRISPR/Cas-based recognition and signaling. Catalytically active Cas proteins and/or deactivated Cas (dCas) will be employed to achieve sequence-specific detection and, where applicable, differentiation of acquired genetic features such as DNA methylation, enabled through recognition strategies and/or tailored amplification. Recombinase polymerase amplification (RPA) or related isothermal methods will be used to generate amplified DNA under instrument-free conditions. Amplified products may support signal generation via alternative pathways, including dCas-mediated capture for binding-based readout or Cas12a-mediated collateral trans-cleavage for signal generation. Detection performance will be assessed using fluorescence-quencher (FQ) reporters and binding measurements, including surface plasmon resonance (SPR), to define analytical sensitivity, specificity, and operational parameters for downstream diagnostic integration.

**Deliverable**: Modular detection system developed.

**Milestone**: Key components optimized and validated for use in LFA platform for at least two biomarkers (from Task 1) in at least one matrix (from Task 1).

***Task 3. System Integration on a Paper-Based LFA Platform (Months 12–24, TRL 3–4)***

The objective of this task is to translate the validated molecular detection system into a paper-based LFA and integrate all functional components into a deployable format. An LFA architecture will be developed that incorporates engineered affinity-based capture chemistries for the immobilization of labeled assay products on the test surface and generate a visible colorimetric signal using gold nanoparticles. Depending on application requirements, assay readout may be performed by naked-eye observation or, where quantitative analysis is desired, by image-based analysis using portable devices such as smartphones. The assay will be engineered to support multiplexed detection of multiple genetic targets on a single strip through spatial or chemical encoding strategies. In addition, sample pretreatment and processing methods compatible with matrices will be developed. These may include simplified nucleic acid extraction, and/or filtration approaches designed for POC or POU usability while preserving assay performance.

**Deliverable**: Provisional patent submitted; joint publication submitted.

**Milestone**: Fully integrated LFA platform established for at least two biomarkers (from Task 1) in at

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| | least one simulated matrix (from Task 1).<br><br>***Task 4. Platform Validation in Simulated Relevant Biological Matrices (Months 25–36, TRL 5)***<br>The final task focuses on validating the fully integrated CRISPR–LFA diagnostic system under relevant, non-clinical conditions consistent with TRL 5 requirements. The detection platform will be evaluated using simulated matrices spiked with target nucleic acids at relevant concentrations. All testing will be conducted *in vitro*. The complete workflow, including sample processing, isothermal amplification, optional CRISPR-based signaling, and LFA-based colorimetric readout will be assessed as an integrated system.<br>**Deliverable**: Korean and U.S. Patent applications filed.<br>**Milestone**: Validated LFA for at least two biomarkers in at least one simulated matrix. |

## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | The research institute (RPI) will be responsible for the development of core CRISPR- and isothermal amplification-based detection technologies. This includes identification and selection of genetic and epigenetic biomarkers, design and optimization of CRISPR/Cas- or dCas-based recognition strategies, integration with isothermal amplification methods suitable for instrument-free operation. In particular, RPI will:<br>1. Develop technologies capable of distinguishing epigenetic modifications, including DNA methylation, using CRISPR- and/or isothermal amplification-based approaches (Tasks 1 and 2).<br>2. Define analytical performance metrics and specifications, including sensitivity, specificity, mismatch discrimination, and epigenetic selectivity (Tasks 2 and 3).<br>3. Collaborate with the Korean SME partner to integrate these molecular detection modules with paper-based LFA formats and to experimentally demonstrate the performance of the fully integrated system using simulated relevant biological matrices (Tasks 3 and 4). |
| **Expected R&R of Korean SME Partner** | The Korean SME partner will lead the development and integration of a paper-based LFA platform incorporating the CRISPR-based detection modules. Core responsibilities include:<br>1. Perform LFA design and engineering, selection of materials and capture chemistry, and development of robust visible readout systems suitable for point-of-care and point-of-use deployment (Tasks 1, 2, and 3).<br>2. Implement multiplexed LFA capability, enabling simultaneous detection of multiple targets on a single strip while minimizing signal cross-reactivity (Task 3).<br>3. Contribute readout enhancement technologies, such as smartphone-based or simple optical analysis, to support qualitative or semi-quantitative interpretation (Task 4).<br>4. Develop and integrate sample pretreatment and processing methods compatible with relevant biological matrices and simplified POC and POU, supporting progress to TRL 5 and future commercialization (Task 4). |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | This CRISPR-driven, paper-based diagnostic platform aims to detect and distinguish epigenetic modifications, such as DNA methylation. Once developed, it could be applied not only to blood but also to minimally invasive samples such as urine and sweat, expanding its potential for patient-friendly testing. By enabling POC testing for clinicians and POU testing for patients, this approach is expected to bring molecular diagnostics closer to both healthcare providers and individuals, supporting timely clinical decisions and frequent, convenient monitoring. Such accessibility benefits patients through reduced invasiveness and faster results, while empowering physicians with actionable information at the bedside. The global point-of-care genetic testing market was valued at approximately USD 6.64 billion in 2024 and is projected to reach USD 7.95 billion by 2031 at a CAGR of around 2.6%, driven by increasing demand for rapid, decentralized genetic diagnostics[15]. This trend highlights the commercial potential for CRISPR-enabled, paper-based assays that combine molecular precision with low-cost, widely deployable formats. By partnering with a Korean SME, the successful development of this platform is expected not only to improve clinical decision-making and expand access to non-invasive diagnostics but also to provide economic and technological benefits through intellectual property generation and potential technology transfer, strengthening the competitiveness of these companies in biotechnology and decentralized molecular testing.<br>**Team**<br>**Jonathan S. Dordick, Ph.D.,** is Institute Professor of Chemical and Biological Engineering at RPI, with a joint appointment in the Department of Biological Sciences. He also serves as Vice President for Strategic Alliances and Translation. He received his B.A. degree in Biochemistry and Chemistry from Brandeis University and his Ph.D. in Biochemical Engineering from the Massachusetts Institute of Technology. At RPI, he has served as the Vice President for Research, founding Co-Director of the Rensselaer-Mount Sinai Center for Engineering and Precision Medicine, Director of the Center for Biotechnology & Interdisciplinary Studies, and Department Chair. Dr. Dordick has made foundational contributions to protein science and enzyme technology, biomanufacturing of therapeutic proteins and nucleic acids, and drug discovery. Dr. Dordick has received numerous awards and honors, including election to the National Academy of Engineering and National Academy of Inventors, and receipt of numerous society awards. He has founded several companies and published over 460 papers and is an inventor/co-inventor on over 50 patents.<br>**Seok-Joon Kwon, Ph.D.** is a Senior Research Scientist at RPI with two decades of expertise bridging molecular biology, protein, engineering, glycobiology, and nanobiotechnology. His research focuses on developing high-throughput screening assays, CRISPR-based systems, and combinatorial methods for drug discovery. He has a portfolio of over 60 publications, including papers in prestigious peer-reviewed journals, book chapters, and patents. |

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## 6. References

(1) Loy, C., Ahmann, L., De Vlaminck, I. & Gu, W. Liquid biopsy based on cell-free DNA and RNA. *Annu. Rev. Biomed. Eng.* **26**, 169–195 (2024).

(2) Pharo, H. et al. From concept to clinic: a roadmap for DNA methylation biomarkers in liquid biopsies. *Oncogene* **44**, 4814–4831 (2025).

(3) Augustine, S., Venkadesh, A., Kaushal, S., Lee, E., Ajaj, M. & Lee, N.-E. Point-of-Care Testing: The Convergence of Innovation and Accessibility in Diagnostics. *Anal. Chem.* **97**, 9569–9599 (2025).

(4) Liu, Z., Zhou, Y., Lu, W. et al. Microfluidic biosensors for biomarker detection in body fluids: a key approach for early cancer diagnosis. *Biomark. Res.* **12**, 153 (2024).

(5) Pedreira-Rincón, J., Rivas, L., Comenge, J., Skouridou, V., Camprubí-Ferrer, D., Muñoz, J., O'Sullivan, C. K., Chamorro-Garcia, A. & Parolo, C. A comprehensive review of competitive lateral flow assays over the past decade. *Lab Chip* **25**, 2578–2608 (2025).

(6) Wu, X., Zha, J., Koffas, M.A.G. & Dordick, J.S. Reducing Staphylococcus aureus resistance to lysostaphin using CRISPR-dCas9. *Biotechnol. Bioeng.* **116**, 3149–3159 (2019).

(7) Gopal, S., Rodrigues, A.L. & Dordick, J.S. Exploiting CRISPR Cas9 in three-dimensional stem cell cultures to model disease. *Front. Bioeng. Biotechnol.* **8**, 692 (2020).

(8) Badri, A., Williams, A., Awofiranye, A., Datta, P., Xia, K., He, W., Fraser, K., Dordick, J.S., Linhardt, R.J. & Koffas, M.A.G. Complete biosynthesis of a sulfated chondroitin in Escherichia coli. *Nat. Commun.* **12**, 1389 (2021).

(9) Lee, I., Kwon, S.J., Sorci, M., Heeger, P.S. & Dordick, J.S. Highly sensitive immuno-CRISPR assay for CXCL9 detection. *Anal. Chem.* **93**, 16528–16534 (2021).

(10) Lee, I., Kwon, S.J., Heeger, P. & Dordick, J.S. Ultrasensitive ImmunoMag-CRISPR lateral flow assay for point-of-care testing of urinary biomarkers. *ACS Sens.* **9**, 92–100 (2024).

(11) Kwon, S.J., Lee, I., Dordick, J.S., Heeger, P. Systems and methods of point-of-care testing for urinary biomarkers including ultrasensitive IMMUNOMAG-CRISPR lateral flow assays PCT/US2024/052198 (2024).

(12) Kwon, P. S., Xu, S., Oh, H., Kwon, S. J., Rodrigues, A. L., Feroz, M., Zhang, F., Hong, J. J., Fraser, K., He, P., Linhardt, R. J. & Dordick, J. S. Suramin binds and inhibits infection of SARS-CoV-2 through both spike protein-heparan sulfate and ACE2 receptor interactions. *Commun. Biol.* **6**, 387 (2023).

(13) Lin, J., Miao, J., Schaefer, K.G., Russell, C.M., Pyron, R.J., Zhang, F., Phan, Q.T., Solis-Swidergall, N.V., Liu, H., Tashiro, M. et al. A genome-scale screen identifies sulfated glycosaminoglycans as pivotal in epithelial cell damage by Candida albicans. *Nat. Microbiol.* **9**, 2553–2569 (2024).

(14) Dean, K. R., Hammamieh, R., Mellon, S. H., Abu-Amara, D., Flory, J. D., Guffanti, G., et. al., Multi-omic biomarker identification and validation for diagnosing warzone-related post-traumatic stress disorder. *Molecular Psychiatry*, **25**, 3337–3349 (2019).

(15) Valuates Reports. *Global Point-of-Care Genetic Testing Market Size, Forecast 2024–2031*.

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## Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
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| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |